

GSEB Class 10 Science

SECTION C - CHAPTER 6

Control and Coordination

3-mark questions | Point-to-point | Exam-oriented

1. What are receptors? State the example, location and function of different receptors in human body.

- Receptors are specialised cells or nerve endings that detect stimuli and receive information.
- Photoreceptors are present in the retina of the eye and detect light.
- Olfactory receptors are present in the nose and detect smell.
- Gustatory receptors are present in the tongue and detect taste.
- Auditory receptors are present in the inner ear and detect sound.
- Skin receptors detect touch, pressure, pain and temperature.

2. Identify the parts of a neuron in the figure. Also answer: (1) Where is information acquired? (2) Through which information travels as an electrical impulse? (3) Where does the impulse get converted into a chemical signal for onward transmission?

- a - Nucleus; b - Cell body (cyton); c - Dendrites; d - Axon; e - Axon terminals (nerve endings).
- Information is acquired at the dendrites of a neuron.
- The information travels through the axon in the form of an electrical impulse.
- At the end of the axon, the impulse reaches the nerve ending.
- At the nerve ending, chemicals called neurotransmitters are released.
- These chemicals cross the synapse and transmit the signal to the next neuron.

3. What is the meaning of reflex actions? Give examples.

- A reflex action is a quick, automatic and involuntary response to a stimulus.
- It occurs without conscious thinking or decision-making.
- Reflex actions are generally coordinated through the spinal cord.
- The pathway followed by the nerve impulse is called a reflex arc.
- Reflex actions help protect the body from sudden harmful stimuli.
- Examples include withdrawing the hand from a hot object, blinking and sneezing.

4. State the organization of human nervous system and general functions of it.

- The human nervous system is divided into the central nervous system (CNS) and peripheral nervous system (PNS).
- The CNS consists of the brain and spinal cord.
- The PNS consists of nerves connecting different body parts with the CNS.
- Receptors receive information from changes in the internal and external environment.
- Nerve impulses carry information to the CNS for processing and coordination.
- The nervous system sends suitable responses and coordinates voluntary and involuntary activities.

5. Describe human brain with specific functions of its different parts. (Diagram is not necessary.)

- The human brain is broadly divided into the forebrain, midbrain and hindbrain.
- The forebrain is associated with thinking, memory, learning, sensations and voluntary actions.
- The midbrain controls certain reflex movements of the eyes, head and neck.
- The hindbrain includes the cerebellum, pons and medulla.
- The cerebellum coordinates voluntary muscular movement and maintains balance and posture.
- The medulla controls involuntary activities such as breathing and heartbeat.

6. For a particular situation which hormone prepares the human body for either fighting or running away from the situation? State the effect of this hormone in animal body.

- The hormone is adrenaline (epinephrine), secreted by the adrenal glands.
- It is released during fear, danger, stress or emergency situations.
- It increases the heart rate so that more blood reaches the muscles.
- It increases the breathing rate and oxygen supply to the body.
- Blood flow to the skeletal muscles increases and digestive activity is reduced.
- It provides quick energy and prepares the body for rapid fight-or-flight action.

7. Explain disorders caused by hormonal imbalance in human beings.

- Dwarfism: deficiency of growth hormone during childhood causes reduced body growth.
- Gigantism: excessive secretion of growth hormone during childhood causes excessive growth.
- Goitre: iodine deficiency may reduce thyroxine production and cause enlargement of the thyroid gland.
- Diabetes mellitus: insufficient insulin or ineffective insulin action causes high blood glucose.
- Hormonal imbalance may disturb normal growth and development of the body.
- It may also disturb metabolism and other important body functions.

8. Observe the following diagram carefully and answer: (1) State the name of a and b with location. (2) State any two functions of b. (3) In which conditions is a stimulated and give the name of its secretion.

- a - Adrenal gland: located on the upper surface of each kidney.
- b - Pancreas: located below and behind the stomach in the abdominal cavity.
- The pancreas produces digestive enzymes that help in digestion of food.
- It also secretes insulin, which helps maintain the blood glucose level.
- The adrenal gland is stimulated during fear, danger, stress and emergency conditions.
- Its important secretion is adrenaline (epinephrine), which prepares the body for rapid action.

9. Give the scientific terms for the following conditions: (1) Bending of shoot towards light. (2) Curling of tendril to nearby objects. (3) The movement of roots towards the water source.

- (1) Phototropism: growth movement of a shoot towards light.
- (2) Thigmotropism: growth movement of a tendril in response to touch or contact.
- (3) Hydrotropism: growth movement of roots towards a source of water.
- These are directional growth movements shown by plants.
- The direction of growth depends on the direction of the external stimulus.
- Thus, the terms are phototropism, thigmotropism and hydrotropism, respectively.

10. Identify me: (1) I am a hormone generally for an emergency, increase breathing rate and change blood flow to digestive system and muscles. (2) I am present in greater concentration at areas of rapid cell division, such as in fruits and seeds. (3) I am controlling your daily activities such as walking, riding a bicycle, picking up objects, etc.

- (1) Adrenaline: an emergency hormone released during danger or stress.
- (2) Cytokinin: a plant hormone associated with regions of active cell division, such as fruits and seeds.
- (3) Cerebellum: the part of the hindbrain that coordinates voluntary muscular activities.
- Adrenaline increases breathing and changes blood flow to prepare the body for rapid action.
- Cytokinin promotes cell division and supports growth in plants.
- The cerebellum also helps maintain balance and posture.

11. State the secretions produced by the following endocrine glands and write their function: (1) Thyroid gland (2) Ovary (3) Pancreas.

- Thyroid gland - Thyroxine: regulates body metabolism and supports normal growth and development.
- Ovary - Oestrogen: controls female reproductive development and secondary sexual characters.
- Ovary - Progesterone: helps regulate the menstrual cycle and supports pregnancy.
- Pancreas - Insulin: helps maintain the normal blood glucose level.
- Pancreas - Glucagon: helps increase blood glucose level when it falls.
- These hormones regulate important processes such as metabolism, reproduction and blood glucose.

12. A father and mother are in an age-group of forty plus. The father has a disturbance in blood sugar level and is advised daily walking and less sugar. The mother is advised a thyroxine tablet daily. Answer the following questions.

- (1) The father's blood test shows high blood glucose or disturbed blood-sugar regulation.
- (2) The pancreas is not functioning properly and the hormone insulin is not produced or released in sufficient amount.
- (3) The gland affected in the mother is the thyroid gland.
- Low thyroxine production causes a decrease in normal metabolic activity.
- Regular walking and a low-sugar diet are advised to help control high blood glucose.
- Thyroxine tablets provide thyroid hormone when the body needs replacement.